

Selected Extracts

Moen Appendix A followed by Yao & Rasmussen Sections 1-2

INTERNAL PAGE CROSS-REFERENCE

Moen, Appendix A (source pp. 418-423)

Resulting PDF pp. 2-7

Yao & Rasmussen, Sections 1-2 (source PDF pp. 7-22; printed pp. 6-21) Resulting PDF pp. 8-23

Citations

1. Moen

Moen, Cristopher Dennis. 2008. Direct Strength Design of Cold-Formed Steel Members with Perforations. PhD dissertation, Johns Hopkins University, Baltimore, Maryland. Appendix A, pp. 418-423.

2. Yao & Rasmussen

Yao, Zhenyu, and Kim JR Rasmussen. 2014. Finite Element Modelling and Parametric Studies of Perforated Thin-Walled Steel Columns. School of Civil Engineering Research Report R948, The University of Sydney. Sections 1-2.

Appendix A

ABAQUS input file generator in Matlab

The finite element models in this thesis were generated with a custom Matlab program which assembles a column or beam with any general cross-section (input in CUFSM-style format) using nine node S9R5 thin-shell finite elements. The user has the ability to add holes at specific locations in the member, dictate the boundary conditions and application of load, specify the material properties, and impose imperfection, residual stresses and plastic strains to define a member's initial state. Input files for eigenbuckling analysis and nonlinear finite element simulations can be generated. The program was used throughout this research to generate groups of ABAQUS input files for parameter studies. The program setup used to generate the nonlinear finite element models of the column experiments is provided here.

```

clear all
close all

sourceloc='C:\Documents and Settings\Cris\Desktop\cmoen\Cold Formed Steel - Holes
Research\Fall 2007\runbuck development\Rev_6NL\jhab'

%This example generates ABAQUS input files for the nonlinear
%simulation of 12 different columns with SSMA C-sections and evenly spaced
%slotted holes. The column boundary conditions are pinned-pinned warping free and the
column is
%loaded from both ends with a uniform compressive stress simulated as
%consistent nodal loads on the first two sets of cross-section nodes.
%25%CDF and 75% CDF imperfections are imposed on the member geometry with
%CUFSM local and distortional buckling mode shapes.

addpath([sourceloc '\functions\filewriting\'])
addpath([sourceloc '\functions\holes\'])
addpath([sourceloc '\functions\'])
addpath([sourceloc '\templates\'])
addpath([sourceloc '\'])

load SSMAxsections %SSMA cross section info
load SSMAnames %SSMA name list
load SSMA_wavelengths %SSMA CFSM wavelengths for Pcr1, Pcrd
load Ag %SSMA gross cross-sectional area

%define the SSMA sections to create models for
sections=[12
86
11
73
39
95
72
56
47
75
66
87]

%define the imperfection magnitudes
imptypes=[25 75]

%define hole length (slotted holes considered here)
Lhole=4
%define rough hole spacing, will be adjusted in holes section of file
S=12

%define member lengths
Lc=[34
88
24
74
42
78
66
56
32
74
40
80]

%define hole depth such that Anet=0.70Ag
Anetfactor=0.7

count=1
for i=1:length(sections)

    section_num=sections(i)

```



```

%IMPERFECTIONS
*****IMPERFECTIONS*****
%type=0    no imperfections
%type=1    use mode shapes from ABAQUS results file
%type=2    input from file
%type 3    impose CUFSM shapes as imperfections

imperfections.type=3;
imperfections.filename=[];
imperfections.step=[];
imperfections.mode=[]

t=dims(15)
if imptypes(j)==25
    if L>24
        imperfections.magnitude=[0.14*t 0.64*t L/2000]
        imperfections.wavelength=[SSMA_wvlenghts(section_num,1)
SSMA_wvlenghts(section_num,2) L]
    else
        imperfections.magnitude=[0.14*t 0.64*t]
        imperfections.wavelength=[SSMA_wvlenghts(section_num,1)
SSMA_wvlenghts(section_num,2)]
    end

elseif imptypes(j)==75
    if L>24
        imperfections.magnitude=[0.66*t 1.55*t L/1000]
        imperfections.wavelength=[SSMA_wvlenghts(section_num,1)
SSMA_wvlenghts(section_num,2) L]
    else
        imperfections.magnitude=[0.66*t 1.55*t]
        imperfections.wavelength=[SSMA_wvlenghts(section_num,1)
SSMA_wvlenghts(section_num,2)]
    end
end
imperfections.plumb=[]
imperfections.member=[1] %1 for column, 2 for beam

%DEFINE HOLES
%Add holes to your member.
%hole.type=1    circular
%hole.type=2    rectangular
%hole.type=3    slotted w/radial ends
%hole.dimension=['width or length (ABAQUS x direction)' 'height or diameter']
%hole.location=['CUFSM cross section node (must be odd!)' 'longitudinal location'
'shift hole in direction of height']
%hole.thickness = thickness of finite elements making up hole, usually the same
as the rest of the member
%I've defined two slotted holes here in the web of the cross-section.

%number of holes
nhole=floor(L/S)

if nhole<1
    nhole=1
end

%final hole spacing
Sfinal=floor(L/nhole)
spacing=Sfinal/2:Sfinal:L-Sfinal/2
hole.type=[3*ones(nhole,1)];
%define dimensions for slotted hole
hole.dimension=[Lhole*ones(nhole,1) hhole*ones(nhole,1)];
%define location of hole in cross-section
hole.location = [(length(node)+1)/2*ones(nhole,1) spacing' zeros(nhole,1)]
hole.thickness = [dims(1,15)*ones(length(hole.type),1)]
hole.material=[100*ones(length(hole.type),1)];
hole.groups=[100000+[1:length(hole.type)]];
hole.fill=[zeros(length(hole.type),1)];
%If you don't want holes, replace above with

```

```

%hole=[ ]

%MEMBER END LOADINGS
%Loading notation is similar to CUFEM. Apply P for compression, M for
%moment, or a combination of both. Compression at both ends of
%column are
%shown here. Loads are applied as consistent nodal loads in ABAQUS.

end1load.P=1;
end1load.Mxx=0;
end1load.Mzz=0;
end1load.M11=0;
end1load.M22=0;

end2load.P=-1;
end2load.Mxx=0;
end2load.Mzz=0;
end2load.M11=0;
end2load.M22=0;

%CALCULATE CONSISTENT NODAL LOADS ON MEMBER ENDS*****
unsymm=0
[end1cload, end2cload, A,
Ixx]=consist_endloads(node,elem,end1load,end2load,unsymm, nL, FSection_increment);

%ABAQUS NODE SETS
%Define these node sets to apply boundary conditions in ABAQUS
%      nodesetinfo=('nodeset name' [xlim1 xlim2 xint] [ylim1 ylim2 yint] [zlim1
zlim2 zint] exclude)
%where nodes are grouped based on xlim1<=x<=xlim2 and ylim1<=y<=ylim2 and
zlim1<=z<=zlim2.
%Instead of ranges, assign xint,yint,zint to something other than zero to group
nodes at specific x,y,and z
%distance intervals
%      xlim1:xint:xlim2, ylim1:yint:ylim2, ylim1:yint:ylim2.
%The exclude command can be used to exclude previously defined node sets from the
current node set.
%exclude = 0    all nodes in range are included in nodeset
%exclude = m    excludes nodeset m from current nodeset

nodesetinfo={'ENDXZERO' [0 0 0] [-1000 1000 0] [-1000 1000 0] 0;
'ENDXL' [L L 0] [-1000 1000 0] [-1000 1000 0] 0;
'DISPDOF' [L L 0] [0 0 0] [0 0 0] 0;
'MID' [L/2 L/2 0] [max(node(:,3))-0.05 max(node(:,3))+0.05 0]
[max(node(:,2))/2-0.05 max(node(:,2))/2+0.05 0] 0};

%DEFINE SPRINGS
springs=[]

%DEFINE CONTACT SURFACES, NODE SURFACES, KINEMATIC CONSTRAINTS,....
surface.type={}
surface.type=[]
surface.local=[]
surface.coord=[]
surface.coupling={};
surface.interaction=[]
surface.contact=[]
surface.areadist=[]

%ABAQUS INP FILE NAME
jobname(count)=[SSMNames(section_num) '_' num2str(i) '_' num2str(imptypes(j))];

%DEFINE ANALYSIS STEP
step(1).stepinfo={'STEP 1,' 'nlgeom,' 'INC=' [260]};
step(1).solutiontype='Static, Riks';
step(1).solutionsteps={'0.25, ,1e-10, 1'};
step(1).solutioncontrols={ };
step(1).boundarycon={'ENDXZERO' 2 3;

```

```

        'ENDXL' 2 3;
        'MID' 1 1)
    step(1).coupling=[]
    step(1).loads={'*Cload' end1cload(:,1) 1 end1cload(:,2)./2;
        '*Cload' end1cload(:,1)+200 1 end1cload(:,2)./2;
        '*Cload' end2cload(:,1) 1 end2cload(:,2)./2;
        '*Cload' end2cload(:,1)-200 1 end2cload(:,2)./2}
    step(1).outrequest={'*Output, field, frequency=10';
        '*Element Output';
        '1,3,5';
        'S,MISES';
        '*Node Output';
        'U';
        '*Node Print, NSET=DISPDOF, SUMMARY=NO';
        'U1,CF1'};

    %WRITE ABAQUS INP FILE
    %this is the important function, you can use this in for loops to generate
parameter studies
    jhabnl(L, node, elem, nele, end1load, end2load, hole, nodesetinfo, surface,
nodeadd, step, jobname(count),matprops,imperfections,springs,sectionpoints)
    count=count+1
end

end

%CREATE BEAST BATCH FILE
%Generates a linux batch file that will submit all of the parameter study
%.inp files to the queue manager on the beast.
ABQbeastscript(jobname,ones(length(jobname),1)*4,'cdmscript')
%Run the script at the beast command line with:
% bash cdmscript

```

1. Introduction

In order to gain a deeper understanding and formulate new design equations for perforated thin-walled steel columns based on the Direct Strength Method (DSM), a comprehensive parametric study was carried out by means of nonlinear finite element modelling in ABAQUS on 146207 perforated and non-perforated column members.

It is vital to first develop a reliable and efficient FE model capable of producing realistic and accurate results, particularly for elastic buckling and non-linear ultimate modes and loads. As the material and geometric non-linear modelling of thin-walled structures is sensitive to modelling inputs and assumptions (Schafer, Li et al. 2010), caution should be exercised when defining issues such as the type and size of the element, the material, boundary conditions, imperfections, and solution controls, etc. The model should also be validated before trusting it to generate further data for design purposes. This report follows these principles and describes the essential stages in the development of FE models for columns predisposed to local, distortional, and global buckling, and any possible interactions between these basic buckling modes. The models were calibrated against documented experimental data and well established numerical and analytical solutions.

Subsequently, by way of a rigorous and systematic procedure, parametric studies were carried out to produce more than one hundred thousand data points. Simulations of perforated columns having five types of cross-sections (i.e. C, stiffened C, Z, rack, and hat sections) and subjected to three major buckling modes (i.e. local, distortional and global buckling), including the interactions between them, were performed. Data obtained from the parametric studies mainly included the elastic buckling loads and ultimate strengths of the FE models, as well as the elastic buckling loads of non-perforated Semi-Analytical Finite Strip (SAFS) models.

ABAQUS/standard v6.10 and CUFSM 3.12 were used as solvers for the FE and SAFS simulations respectively, while Matlab codes were developed to perform most of the pre-processing (including generating input files) and post-processing tasks. Simulations were performed mostly on (i) the Sun Constellation VAYU Supercomputing Cluster provided by Australian National Computational Infrastructure (NCI) Facility (n.d.), and (ii) three DELL PowerEdge R910 rack servers (two servers running on Rocks 5.4 based on the CentOS Linux and one server running on Microsoft Windows Server 2008 R2 Enterprise), each with four E7540 six-core Intel Xeon processors and 64 GB RAM.

2. Development of FE models

2.1. Element and meshing

2.1.1. Element type

It is widely acknowledged that for thin-walled structures, shell elements are the most suitable type for FE modelling because it is able to capture local, distortional, as well as localised buckling of the structure so that a realistic response can be predicted. ABAQUS offers a dozen commonly used shell elements for 3-D stress/displacement analysis, such as S3, S3R, S4, S4R, S4R5, S8R, S8R5, S9R5, etc., of which S9R5, as shown in Fig. 1, was chosen for this study. In short, S9R5 is a 9-node doubly curved thin shell element with reduced integration (four integration points) that uses five degrees of freedom per node (three displacements and two in-plane rotations), and is applicable to large rotations but small strains. The flow chart shown in Fig. 2 details the element selection process based on the information provided in the ABAQUS Manual (ABAQUS/STANDARD 2010) and the specific demands of the present study.

In addition, Moen (2008) demonstrated the merits of element S9R5 after conducting a series of numerical experiments on single plates, by comparing the relative performance between elements S4, S4R, and S9R5. It was shown that because S9R5 used a quadratic shape function rather than linear shape functions as used for S4 and S4R, it significantly outperformed S4 and S4R in that it required much fewer nodes to achieve the same level of accuracy.

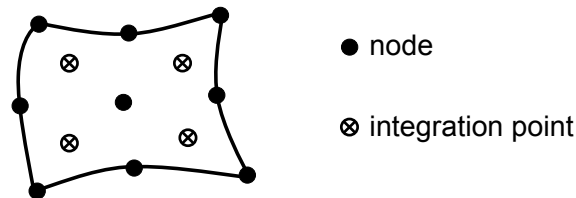
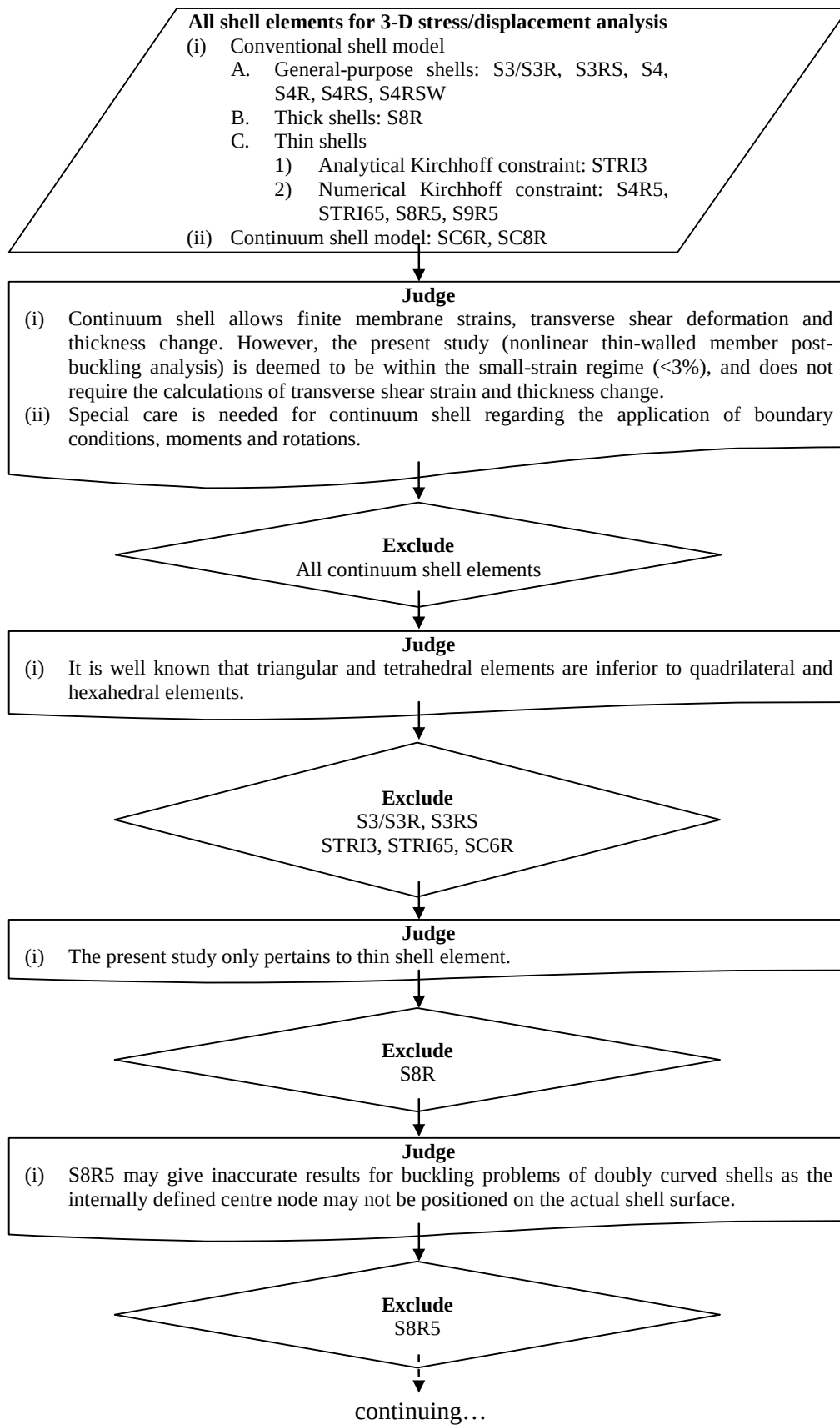


Fig. 1: Illustration of ABAQUS shell element S9R5



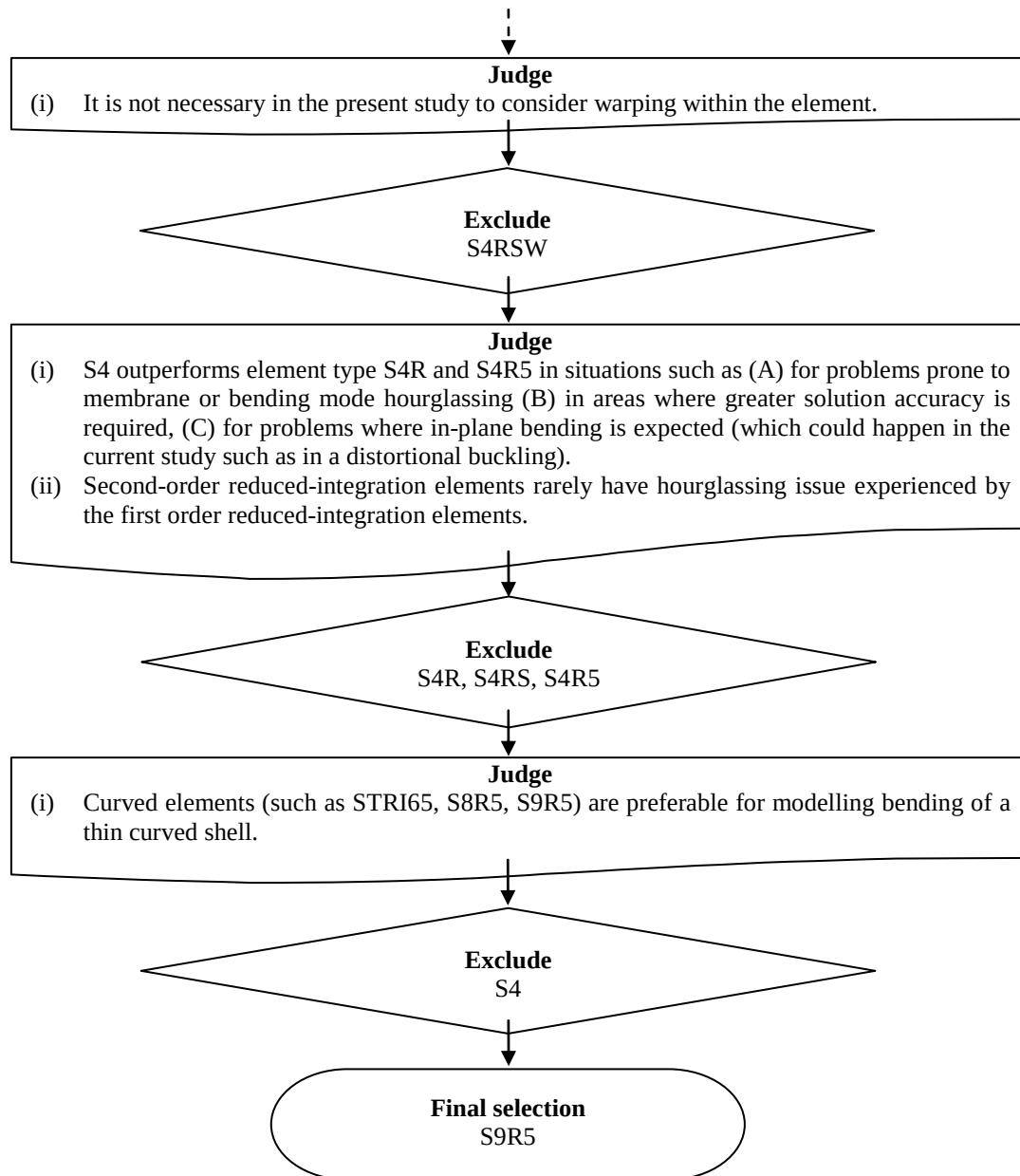


Fig. 2: Element selection process

2.1.2. Element mesh

The present study imposed the following overall restraints on the meshing algorithm of the FE simulations in order to optimise the balance between accuracy and computational efficiency:

- (i) An overall (or average) mesh size (i.e. seed size (SS)) was established to control the mesh size wherever possible. It is defined as:
 $SS = \text{minimum of } (SSA, SSB, SSC, SSD, SSE)$, where
 $SSA = 1/4$ of overall web width,
 $SSB = 1/4$ of overall flange width,
 $SSC = 1/4$ of local buckle half-wavelength,

$SSD = 4 \cdot B_{\text{EleSide}}$, where B_{EleSide} is the typical mesh size at the edges of holes, as shown in Fig. 5,

$SSE = 1/20$ of member length for members predisposed to global buckling.

- (ii) Two elements were provided per rounded corner in the cross-section with the maximum element aspect ratio of 16 (Moen 2008).
- (iii) The aspect ratios for rectangular flat elements were kept below 8 (Moen 2008).

However, the meshing criteria (i-iii) were only sufficient for non-perforated members. For perforated members, the meshes and their quality in areas near the holes had a profound impact on issues such as the overall mesh quality and density of the member, the accuracy of the results, and the tendency towards convergence difficulties due to excessively distorted elements. It was considered in the present work that the proper meshing of perforated members should also involve the following principles:

- (iv) Meshes around holes should be denser than in areas away from holes and should be dense enough to realistically simulate the response of these regions (such as localised buckling around holes), although not too dense so as to impair computational efficiency.
- (v) Meshes around holes should have as regular as possible shapes and patterns so as to reduce element distortion, possible inaccuracy in the results, and convergence difficulties during analysis.
- (vi) Mesh checking and re-meshing algorithms should be available in order to ensure that every element in the member satisfies the overall meshing restraints (i-iii), and make adjustments if needed.

The Matlab input file generator used in this study was able to create four types of holes, i.e. circular, slotted, square, and rectangular, as shown in Fig. 3. The slotted and rectangular holes can have two different orientations in a column, i.e. with their longer dimension parallel or perpendicular to the length of the column, and each orientation required a unique meshing algorithm.

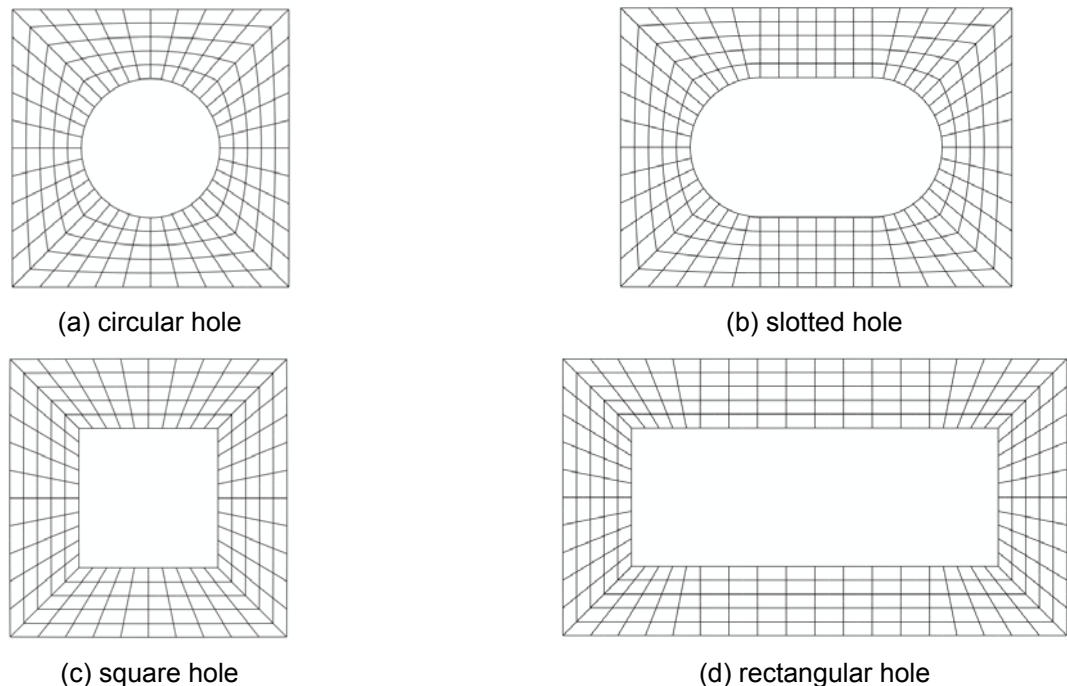


Fig. 3: FE meshing of the hole region for different hole shapes

Depending on the specific geometries of the width, length, and spacing of the hole, as well as the length and cross-sectional geometry of the column, the meshing algorithm used in this study was iterative and adaptive in order to satisfy all the meshing criteria listed above. The general meshing procedure is shown in Fig. 4. It first generated meshes near the hole and then in the remaining (base) structure, and it finally completed the mesh by joining the two basic mesh components.

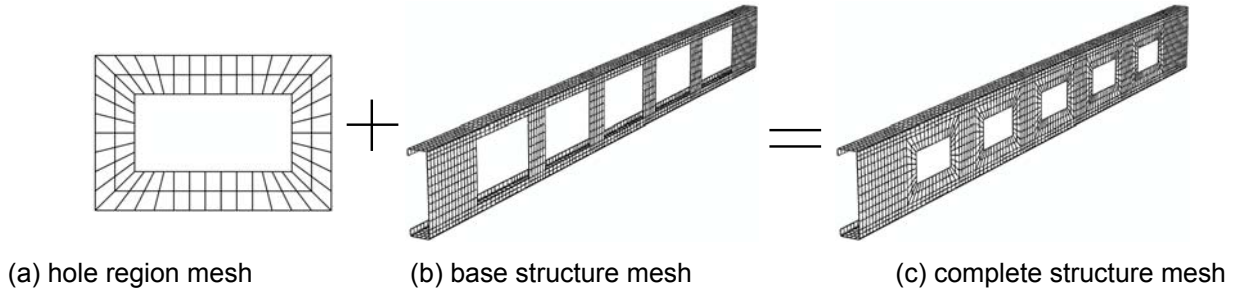


Fig. 4: General meshing procedure

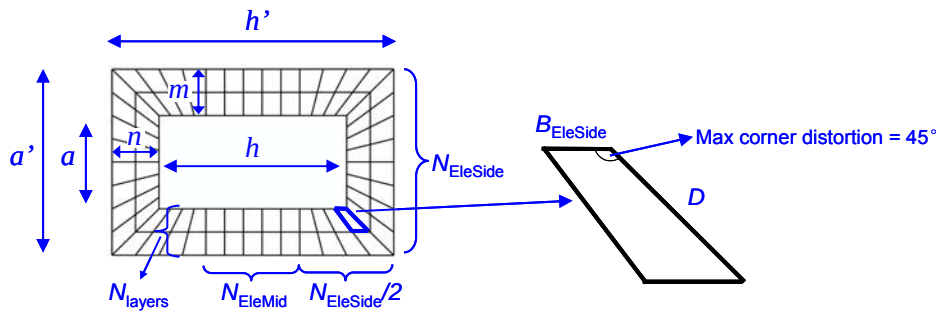


Fig. 5: Hole region meshing

More specifically, by taking the meshing in Fig. 4 as an example of a column with deep rectangular holes, the general meshing procedure is broken down as follows:

- (i) Decide on the width of the hole region (a') as $a' = \min(2 \times a, b)$, where a is the hole width and b is the flat width of the C-section web.
- (ii) Decide on the length of the hole region (h'). Note that ABAQUS suggests that distortion in the corner angles of an element should not be more than 45 degrees. This requires that the distances m and n in Fig. 5 should be equal which results in $h' = a' + h - a$.
- (iii) Check if h' is larger than the centre to centre spacing of the hole (s'), and if the answer is yes, scale down h' and a' so that $h' = s'$ and $a' = h' - h + a$.
- (iv) Prescribe the number of elements (N_{EleSide}) along the short edge of the hole. In order to accurately model the stress concentration around the corners of the hole, $N_{\text{EleSide}} = 6-10$ is recommended, with the width of each element equal to $B_{\text{EleSide}} = a / N_{\text{EleSide}}$. Note that a larger number of N_{EleSide} could result in an overly dense mesh and thus increase the cost of computation. However, members subjected to local buckling, and large holes subjected to localised buckling usually require denser meshes around the holes, so 10 may be used for N_{EleSide} .
- (v) Determine the number of elements along the long edge of the hole. As shown in Fig. 5, the long edge consists of three segments of elements, with two identical segments on the sides (each has

$N_{\text{EleSide}}/2$ elements with the width B_{EleSide}) and a middle segment. Considering that stress concentration usually occurs in the corner areas, the middle segment does not require the meshes to be as dense as those in the corner areas. Therefore, the size of the mesh in the middle segment is defined as the average of the seed size and B_{EleSide} , so that the element number in the mid segment is $N_{\text{EleMid}} = \text{round}((h-a)/((SS+B_{\text{EleSide}})/2))$ with the width of each element determined as $B_{\text{EleMid}} = (h-a)/N_{\text{EleMid}}$, where “round” means rounding to the nearest integer.

- (vi) Hitherto, the circumferential meshes have been defined for the hole region, leaving only the radial meshes, i.e. the number of element layers to be defined. According to Moen (2008), the aspect ratio ($B_{\text{EleSide}}:D$) of the corner elements, as shown in Fig. 5, must stay between 0.5 and 2.0 in order to maintain an acceptable level of accuracy. The current scheme keeps the ratio of $B_{\text{EleSide}}:D$ between 0.5 and 2.0 but close to 0.5 so as to reduce the number of element layers and improve computing speed. The number of element layers is therefore calculated as $N_{\text{layers}} = \text{ceiling}((h'-h)/(2.828 \times B_{\text{EleSide}}))$, where “ceiling” means rounding up to the next integer. In this way the resulting aspect ratio of all the elements in the hole region will be in the range of 0.5-0.75.
- (vii) Once the meshes within the hole regions have been defined the remaining base structure can then be meshed. Care should be taken to ensure that the nodal positions along the perimeters of the hole regions are compatible with those of the base structure. The complete meshes are then generated by patching the hole regions into the base structure by appending the nodal and elemental information to the input file.

2.2. Material modelling

2.2.1. Material properties of the flats

As this study pertains to thin-walled steel structures, von Mises yield criterion and isotropic strain hardening were used to define the yield surface and loading surface, while the associated flow rule was used to calculate the plastic strains.

Thin-walled steel is commonly cold-formed, which means that its uniaxial stress-strain curve is usually different from that of hot-rolled steel. Instead of having a sharp yield point and yield plateau, the stress-strain curve of thin-walled steel usually features a rounded transition from the elastic to the plastic region, with continuous strain hardening.

For the parametric studies, the widely used Ramberg-Osgood stress-strain curve with different n values ($n=5, 10$, and 20) was adopted, it is expressed as

$$\varepsilon = \frac{\sigma}{E_0} + 0.002 \left(\frac{\sigma}{\sigma_{0.2}} \right)^n \quad (1)$$

where E_0 is the initial Young's modulus, $\sigma_{0.2}$ is the so-called proof stress corresponding to a plastic strain of 0.2%, and n is a parameter used to define the sharpness of the knee of the stress-strain curve.

However, the material properties obtained from experimental uniaxial coupon tests were used to validate the FE models against the experiments. It is worth noting that the engineering stress-strain data must be converted to true stress-strain data before inputting into ABAQUS as a series of data points. The conversion formula can be found in (Yao and Rasmussen 2011)

2.2.2. Material properties of the corners

The corners of thin-walled steel sections usually exhibit higher yield stress and ultimate tensile strength because of the cold forming process, especially when the ratios between virgin ultimate tensile strength and virgin yield stress are high, and low between the corner radius and thickness. This study considered the strength enhancement in the corners by using the formulae in Clause 1.5.1.2 of AS/NZS 4600 to calculate the enhanced yield stress σ_{yc} , such that:

$$\sigma_{yc} = \frac{B_c \sigma_{yv}}{(\gamma_i/t)^m} \quad (2)$$

where

$$B_c = 3.69(\sigma_{uv} / \sigma_{yv}) - 0.819(\sigma_{uv} / \sigma_{yv})^2 - 1.79 \quad (3)$$

$$m = 0.192(\sigma_{uv} / \sigma_{yv}) - 0.068 \quad (4)$$

σ_{yv} = virgin yield stress

σ_{uv} = virgin ultimate tensile strength

γ_i = inside corner radius

t = thickness

Although it is recommended in AS/NZS 4600 that Equation (2) be used only when $\sigma_{uv} / \sigma_{yv} \geq 1.2$ and $\gamma_i/t \leq 7$, it was used for all the sections in the parametric studies presented in this report with a constraint $\sigma_{yc} \geq \sigma_{yv}$ imposed for the cases where equation (2) produced lower values than σ_{yv} .

2.3. Boundary conditions, column length, restraints and loading

Boundary conditions have a profound impact on the structural response and therefore they must be modelled as closely as possible to the physical situation in practical thin-walled steel systems. Despite this, their numerical models should not be overly complicated so as to impair modelling efficiency and computational stability.

The boundary conditions, cross-sectional corner restraints, and loading are detailed in Fig. 6, Fig. 7 and Fig. 8 for columns predisposed to local, distortional, and global buckling failures, respectively.

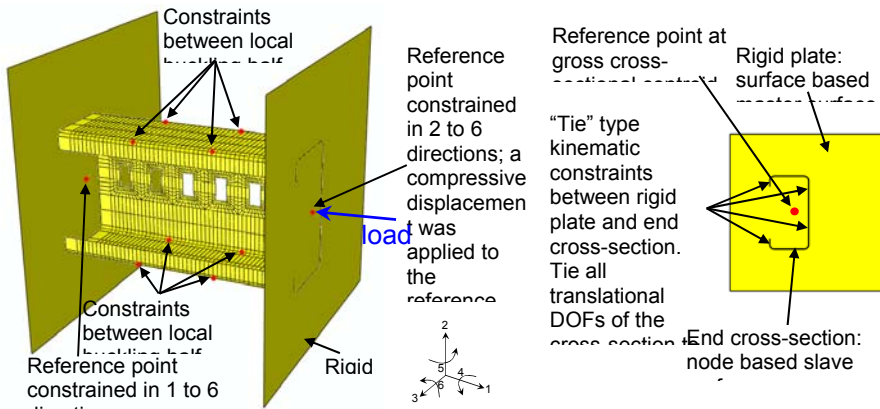


Fig. 6: Boundary conditions, restraints, and loading for columns predisposed to local modes

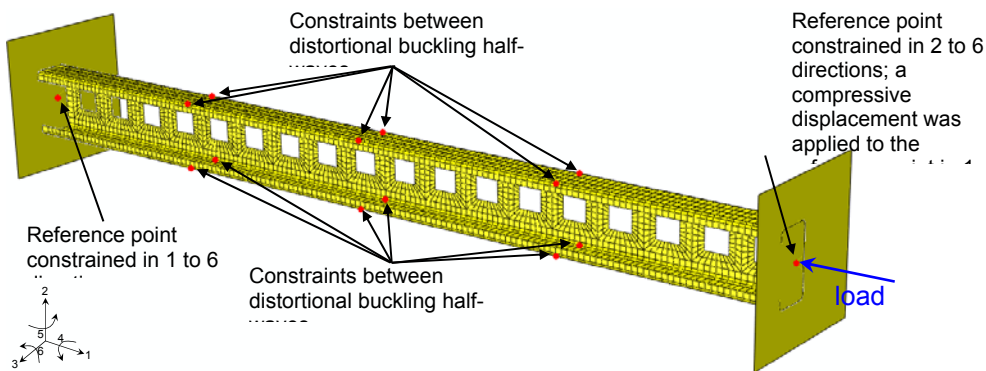


Fig. 7: Boundary conditions, restraints, and loading for columns predisposed to distortional modes

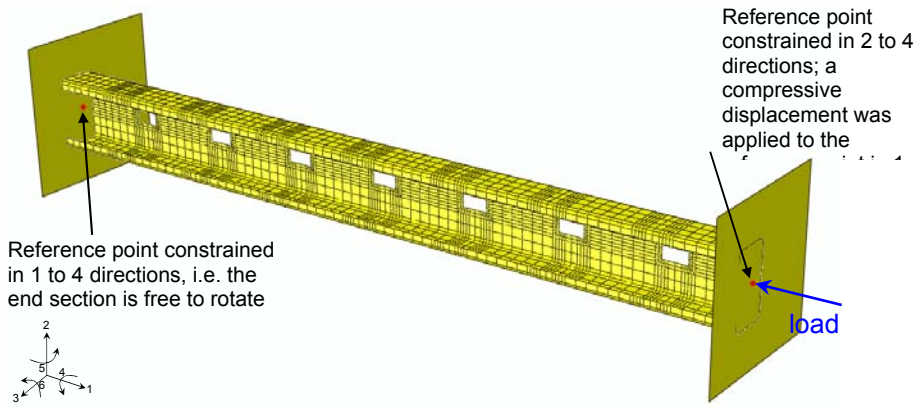


Fig. 8: Boundary conditions, restraints, and loading for columns predisposed to global modes

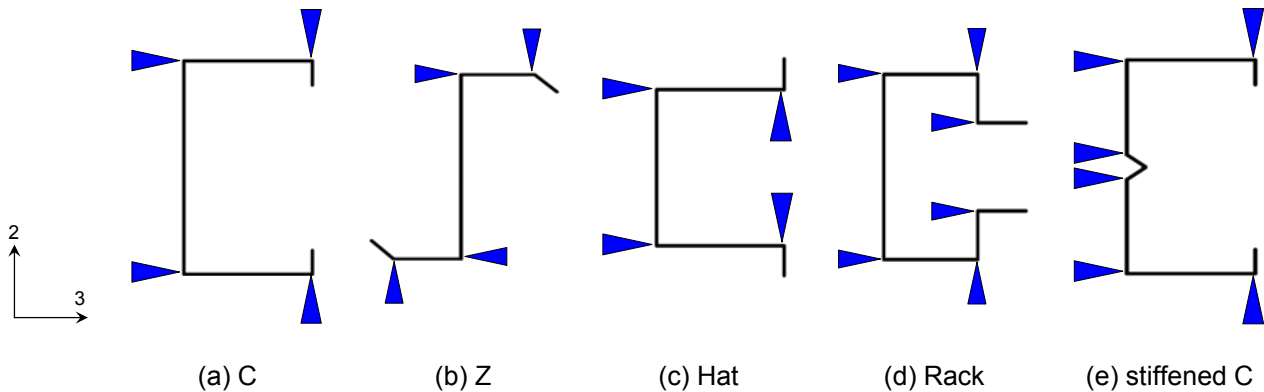


Fig. 9: Illustrations of cross-sectional corner restraints between buckling half-waves

The generic column model used in this study comprised a certain number of buckling half-wavelengths, i.e. three for local, four for distortional, and one for global failures, with the half-wavelengths calculated by the Semi-Analytical Finite Strip Method (SAFSM) based on the non-perforated gross cross-section. Multiple half-wavelengths were chosen for local and distortional failures because (i) the end effects during loading can be minimised, especially in distortional buckling where significant warping displacement is present (see Section 3.1 and 3.2 for detailed discussions on the effect of member length), (ii) thin-walled columns in practice usually buckle in multiple half-waves along the length, (iii) the columns need to be long enough to be able to accommodate a variety of hole patterns in the parametric studies, (iv) a length of three local buckling half-waves for columns failing in local modes is consistent with the lengths of the specimens in most documented laboratory tests on stub columns, as well as the length of specimen suggested in Clause 8.1.2 of AISI S902-08 for stub-column test method of cold-formed steel columns (American Iron and Steel Institute (AISI) 2008), (v) a length of four distortional buckling half-waves for columns predisposed to distortional failure is considered long enough to minimise the end effects (see 3.2 for discussions), and is also consistent with the

minimum length of the specimen recommended in Clause 7.1 of AISI S910-08 for distortional buckling tests of cold-formed steel hat shaped columns (American Iron and Steel Institute (AISI) 2008).

Corner restraining systems were provided for the columns with lengths consisting of multiple buckling half-waves (i.e. columns predisposed to local and distortional failures). Therefore, in the columns failing in a local mode, distortional and global deformation would be excluded, while in columns predisposed to distortional failure, global mode deformation would be prevented. Each set of corner restraints was placed between the buckling half-waves, as shown in Fig. 6 and Fig. 7. The details regarding the restrained DOFs for each set of corner restraints are shown in Fig. 9 for the cross-sections included in the parametric studies. If the C-section in Fig. 9(a) is used as an example, the web-flange and flange-lip corners were restrained in the 3 and 2 directions, respectively. It should be noted that the columns predisposed to local and distortional failures shared the same restraining schemes as shown in Fig. 9, which indicates that the restraints were also placed at the flange-lip corners of the columns predisposed to distortional buckling. This was because (i) the restraints were placed only at the cross-sections between the buckle half-waves, which, in theory, were not deformed during loading (except for negligible Poisson expansions), and (ii) a consistent modelling was ensured by maintaining the specified distortional buckling half-wavelength. In addition, the same restraining schemes were also applied to the perforated members.

By means of these corner restraining systems, (i) a column as shown in Fig. 6 would only fail in local and/or localised mode(s), (ii) a column as shown in Fig. 7 was free to fail by interaction between localised, local and distortional modes, and (iii) a column as shown in Fig. 8 was free to fail by interaction between localised, local, distortional and global modes.

Both ends of all the columns were supported on rigid plates. The end cross-section and rigid plate were connected by "Tie" type kinematic constraints such that all the translational DOFs of the end cross-section were tied to those of the plate, while all the rotational DOFs of the end cross-section remained independent. A reference point was created within each rigid plate whose position was coincident with the centroid of the gross cross-section. Support conditions and loading were applied to the column via the reference point. Apart from an axial load that was applied to the reference point of the top rigid plate as a compressive displacement in the 1 direction, all the translational and rotational DOFs (i.e. 1-6) were restrained for the columns predisposed to local and distortional failures in order to simulate overall fixed-end conditions, while all the translational DOFs (i.e. 1-3) and the rotational DOF about the length of the column (i.e. 4) were restrained for those columns predisposed to global failure in order to simulate overall pinned-ended conditions.

In a non-linear analysis, those columns with a length of multiple half-waves supported on rigid plates and loaded by a compressive displacement, are considered to be superior to those adopting a column length of a single buckle half-wavelength and using free-to-warp boundary conditions with uniform stress applied at the column ends. This is because the latter is unable to replicate the stress re-distribution in the cross-section in the post-buckling stage, and will thus overstress the flexible parts of the buckled cross-section and result in overly conservative ultimate load values.

2.4. Initial geometric imperfections

In practice, a geometrically perfect structure is non-existent, and the non-linear response and ultimate strength of thin-walled steel columns are generally sensitive to initial geometric imperfections. It is therefore essential to include a proper initial geometric imperfection into the numerical model to simulate the actual behaviour of the structure. Although including measured geometric imperfections in the modelling is an ideal approach in terms of accurate results, its application is usually restricted to the validation of numerical models against experimental data. This approach is not feasible for parametric studies since measured imperfection data are mostly not available for simulated specimens. Attempts have recently been made in (Schafer and Pekoz 1998; Schafer, Li et al. 2010) to base the distribution of imperfections, mostly magnitudes, on measured data, so statistical information has been provided for use in the development of numerical modelling and design criteria. This study therefore decided to use the linear elastic buckling mode as the shape of the geometric imperfection and scale it to a magnitude corresponding to an expected level of probability of exceedance.

These local and distortional imperfection shapes (including modes and half-wavelengths) were calculated by the SAFSM linear elastic buckling analyses, and they normally corresponded to the minima in the load vs. half-wavelength curve (signature curve). However, if a minimum was not present, an analysis by the constrained Finite Strip Method (cFSM) (Adany and Schafer 2006), which constrained the mode to either pure local or distortional mode, was used to find the half-wavelength that corresponded to the minimum of the signature curve for the pure mode, and then another general SAFSM analysis was run at that half-wavelength to obtain the corresponding unconstrained mode which was then used as the basic imperfection shape. Local and distortional imperfections were imposed on the column such that they were symmetrical about the mid-length of those columns predisposed to local and global failures, and anti-symmetrical about the mid-length of the columns predisposed to distortional failure. Regarding the shape of the global imperfection, a single sinusoidal wave was assumed along the length of the column and the cross-sectional mode was calculated by a SAFSM analysis performed at the length of the column. If a pure global mode was not present, a cFSM analysis was performed by constraining the model to the global mode.

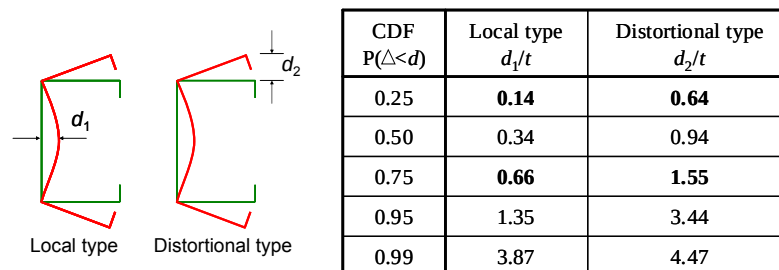


Fig. 10: CDF values for maximum local and distortional imperfections (Schafer and Pekoz 1998)

For the imperfection magnitudes, as per Schafer and Pekoz (1998), the values of two probabilities of exceedance were used as the basis of the magnitudes of local and distortional imperfections, i.e., the 25th and 75th percentile values of the CDF (cumulative distribution function) respectively, as shown in Fig. 10.

Meanwhile, two magnitudes of global imperfections were selected for the parametric studies, i.e. $L/2000$ and $L/1000$ where L is the column length. These two values were used in conjunction with the 25th and 75th percentile magnitudes, respectively. The use of $L/1000$ as the magnitude of the global imperfection for cold-formed steel members is related to the maximum out-of-straightness value specified in many specifications for steel structures. $L/1000$ is usually considered as constituting an upper limit of the measured global imperfection of the column. (Note that recent work by Zeinoddini-Meimand (2011) suggested that the two values of magnitude ($L/2000$, $L/1000$) adopted for global geometric imperfections were conservative in terms of their corresponding percentile values based on measured data. In particular, the newly obtained 25th and 75th percentile values of the bow-shape geometric imperfection were reported as $L/4755$ and $L/1659$ respectively.)

Moreover, the direction of the initial geometric imperfection was explicitly considered in the FE modelling of this study. It was widely recognised that the magnitude and shape were two important parameters for the initial geometric imperfections, while little attention has been paid in most previous works to the direction of geometric imperfection in modelling thin-walled structures. However, it has been found that the direction of imperfection could have a profound impact on the structural response and hence the ultimate strength, as shown by Dinis and Camotim (2007; 2011), and as also demonstrated in (Yao and Rasmussen 2012) and later, in Section 3.1.

Table 1: Imperfection scenarios considered in parametric studies

Predisposed buckling type	Imperfection shape	Imperfection magnitude	Imperfection direction	Resulting imperfection scenarios
Local	Local	A: 25 th * B: 75 th ♣	□: "inward" □: "outward"	A□, A□, B□, B□
Distortional	Distortional	C: 25 th D: 75 th	□: need not be specified	C□, D□
Global	Local	E: $\frac{\sqrt{2}}{2} \cdot 25^{\text{th}}$ F: $\frac{\sqrt{2}}{2} \cdot 75^{\text{th}}$	□: "inward" □: "outward"	If F mode, EGI□□□ EGI□□□ EGI□□□ EGI□□□ FHJ□□□ FHJ□□□ FHJ□□□ FHJ□□□ If FT or T mode, EGI□□□ EGI□□□ FHJ□□□ FHJ□□□
		G: $\frac{\sqrt{2}}{2} \cdot 25^{\text{th}}$ H: $\frac{\sqrt{2}}{2} \cdot 75^{\text{th}}$	□: inward □: outward	
	Global	I: 2000/L J: 1000/L	If F mode ☆, □: toward web □: toward flange	
			If FT or T mode ♦, □: need not be specified	

* 25th: 25th Percentile CDF magnitude

♣ 75th: 75th Percentile CDF magnitude

☆ F mode: flexural mode from a linear elastic SAFSM buckling analysis

♦ FT or T mode: flexural-torsional or torsional mode from a linear elastic SAFSM buckling analysis

The specific scenarios for the geometric imperfection considered in the parametric studies are listed in Table 1, and described as follows:

- (i) For columns predisposed to local failure (e.g. Fig. 6), “inward” and “outward” (see Fig. 11 for the definition of these directions) local buckling modes with the 25th and 75th percentile local-type magnitudes were considered, with the distribution of local imperfection being symmetrical about the mid-length of the column.
- (ii) For columns predisposed to distortional failure (e.g. Fig. 7), distortional buckling modes of an arbitrary direction (inward or outward) with the 25th and 75th percentile distortional-type magnitudes were considered. The direction was arbitrary because any change in the direction of the imperfection would not change the effect due to the anti-symmetry of the distribution of the imperfection about the mid-length of the column. This indicates that a column predisposed to distortional buckling would fail inwards or outwards (whichever is weaker), and thus always fail in the critical direction.
- (iii) For columns predisposed to global failure (e.g. Fig. 8), all the three types of imperfection shapes (i.e. local, distortional, and global mode) were considered and superimposed onto the column with the same level of magnitudes (i.e. either 25th+L/2000 or 75th+L/1000). Their distributions were symmetrical about the mid-length of the column and their maximum values coincided at the mid-length.

Because both local and distortional buckling modes contributed to the local-type displacement shown by d_1 in Fig. 10, to ensure that the magnitude of the final superimposed imperfection was consistent with the percentile level of the individual imperfections, the magnitudes of the individual imperfections were normalised by enforcing the following condition:

$$\left(\frac{d_1}{d_1(P^{\text{th}} \text{ CDF})} \right)^2 + \left(\frac{d_2}{d_2(P^{\text{th}} \text{ CDF})} \right)^2 = 1 \quad (5)$$

where $d_1(P^{\text{th}} \text{ CDF})$ and $d_2(P^{\text{th}} \text{ CDF})$ denote the P^{th} percentile local- and distortional-type imperfection magnitudes, respectively. Furthermore, equation (5) can be represented by a circle in the 2-D plane, as shown Fig. 12. In order to ensure an equal participation by both the local- and distortional-type imperfections, the combined imperfection shape adopted in this study lay on the circle and corresponded to angles where $\theta = 45^\circ$ and 225° , assuming that d_1 and d_2 of the same sign represent the same direction of imperfection, i.e. both inwards or outwards. Therefore, the magnitudes of local and distortional imperfections listed in Table 1 must be scaled by the value of $\sin(\theta = 45^\circ) = |\sin(\theta = 225^\circ)| = \sqrt{2} / 2$.

With regards to the global imperfection component, if the buckling mode calculated by the SAFSM appeared as a flexural mode, two imperfection directions were considered, i.e. with the bow towards the web or the flanges, whereas if it was a flexural-torsional or torsional mode, an arbitrary direction of imperfection could be used because it would make no difference to the strength of the column.

Consequently, as shown in Table 1, those columns predisposed to local, distortional, and global failures could have 4, 2, and 8 (or 4) imperfection scenarios respectively. Note that the perforated columns shared the same distributions of geometric imperfection as the non-perforated columns having the same overall dimensions.



Fig. 11: Definition of sign of local buckles

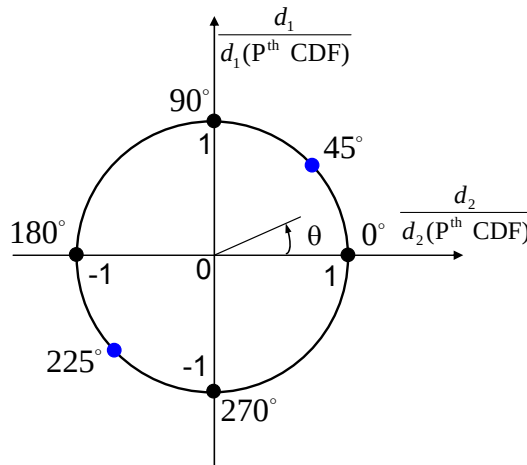


Fig. 12: Representation of superimposed local (d_1) and distortional (d_2) imperfections in a 2-D plane

2.5. Initial residual stresses and plastic strains

Cold-forming during manufacturing imparts residual stresses and plastic strains to the cold-formed steel members that can influence their structural behaviour and ultimate strength to a certain extent. By modelling the actual manufacturing process, analytical models for predicting residual stresses and plastic strains were recently developed by Moen (2008). These models considered (i) the longitudinal and transverse residual stresses, and (ii) the effective plastic strains resulting from coiling, uncoiling, and flattening the sheet, as well as roll-forming. The initial residual stresses and plastic strains from these models can be readily incorporated into ABAQUS, although they were not considered in this research because (i) researchers (Gardner and Nethercot 2004; Wanniarachchi 2005; Becque 2008; Moen 2008) generally agree that their inclusion has only minimal influence on the ultimate load-carrying capacity of thin-walled structures, and (ii) the characteristic of

gradual yielding in the material stress-strain curve for cold-formed steel (such as the Ramberg-Osgood curve used in this parametric study) partially incorporates the loss in stiffness due to initial residual stresses.

2.6. Non-linear solution method and control

In a non-linear FE analysis, the nature of searching for converged solutions at discrete points along the equilibrium path, as well as the global and/or local instability inherent in the FE model, requires a robust non-linear solver and validated control parameters. The arc-length method and the Newtown-Raphson method with applied displacement are the two commonly used non-linear solution schemes. The former is based on the Newtown-Raphson method and uses an automatic arc length constraint in the load-displacement space (see (Yao and Rasmussen 2011) for more details). It is able to trace highly non-linear and unstable load-displacement responses at the global level, and should typically be used when snap-through and snap-back behaviour is expected in the load-displacement equilibrium path, or when the application of a load (instead of displacement) is required in the analysis. Nevertheless, the arc-length method occasionally miscalculates the loading direction and reverts back along the equilibrium path, which is explained by the ABAQUS Theory Manual (ABAQUS/STANDARD 2010) as being the result of very high curvature in the load-displacement space. Moreover, the arc-length method implemented in ABAQUS is unable to include a stabilisation algorithm (or viscous damping) in the solution, but damping is normally required when localised instability such as localised buckling occurs. Therefore, this study adopted the Newtown-Raphson method as the non-linear solver, in conjunction with the application of displacement as the loading and viscous damping for dealing with localised instability. Compared to the arc-length method, this scheme was more stable and transparent, and was sufficient for models where displacement was applied as loading, and snap-through and snap-back issues were not present. More comparisons between these two solvers can also be found in (Moen 2008) and (Schafer, Li et al. 2010).

On the other hand, the solution control methods also play an important role in ensuring an accurate result in a non-linear FE analysis. In order to establish a reliable modelling protocol, a dedicated effort has been made in this research and is presented in detail in APPENDIX A to study the effects of different damping schemes, different magnitudes of “adaptive damping”, the initial step size, maximum step size, and total applied displacement. Although the effect of damping was studied by Moen (2008), the discussions were mostly limited to plates and a damping scheme with a constant damping factor. In addition, no literature found so far contains a study of the effects of the initial step size, maximum step size, and total applied displacement. Overall, the investigations presented in APPENDIX A demonstrate that the aforementioned control parameters all have a certain level of influence on the accuracy of the ultimate load of a FE model of a column, and this level of influence usually varies under those conditions defined by the choice of damping schemes, the magnitude of the control parameters, the failure mode of the column, the cross-sectional slenderness, and the length of the column, etc. In particular, the following tentative conclusions have been drawn: (i) the adaptive automatic stabilisation scheme was much superior to the scheme of automatic stabilisation with a constant damping factor, and was thus used in this research, (ii) the accuracy tolerance used in the adaptive